

CoFiTok: Coarse-to-Fine Denoising Tokens for Pixel-Space Diffusion

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Abstract

Pixel-space diffusion models usually predict a full dense noise field with a single monolithic network output. We introduce CoFiTok, a denoising-token factorization that represents the dense noise prediction as an ordered sum of negative-noise components. Each token is expanded by a restricted, condition-free synthesis operator that only sees the current token, so prefix sums form valid partial noise predictions and produce controllable partial denoising results in pixel space. On CIFAR-10, Tiny ImageNet-200, and an ImageNet-1K 64x64 fallback, a tiny CoFiTok predictor with restricted synthesis learns ordered components with clean zero-token diagnostics, strong random/reverse-order sensitivity, and broader token utilization than endpoint-only factorized training. At the same 20k-step budget, K8 CoFiTok nearly matches endpoint-only final clean MSE on Tiny ImageNet-200 and ImageNet-64 HF while improving prefix path AUC by roughly 28–35x. Current generated-sample Frechet and official FID metrics are mixed, so the supported claim is prefix-controllable denoising at comparable endpoint quality rather than unconditional generation-quality wins.

1 Introduction

Diffusion models reverse a noising process by repeatedly predicting a dense noise field. This dense prediction is effective, but it is also monolithic: there is no direct way to ask how much of the denoising update has been explained, whether the update has an interpretable coarse-to-fine order, or how to stop after a valid partial prediction. Ordered visual tokenizers and diffusion-autoregressive methods suggest that generation can benefit from structured intermediate representations. The question in this paper is narrower: can the dense pixel-space noise prediction itself be factorized into ordered denoising tokens?

CoFiTok rewrites the predicted noise field as

$$\hat{\epsilon}^{(K)} = \sum_{k=1}^K S_k \left(z^{(k)} \right), \quad (1)$$

where each $z^{(k)}$ is a denoising token and S_k expands only that token into one dense component. Prefixes are meaningful by construction:

$$\hat{\epsilon}^{(1:m)} = \sum_{k=1}^m S_k \left(z^{(k)} \right), \quad (2)$$

$$\hat{x}_0^{(m)} = \frac{x_t - \sigma_t \hat{\epsilon}^{(1:m)}}{\alpha_t}. \quad (3)$$

The central design constraint is that the synthesis operator must not become a decoder. In the default CoFiTok model, S_k is token-only, bias-free, shallow, local, and condition-free. It does not

receive the noised image, timestep, class label, text prompt, previous tokens, skip features, or a learned constant. This restriction forces the token itself to carry the sample-specific negative-noise information and gives the zero-token test a concrete meaning: $S_k(0) = 0$ by construction.

The current evidence supports a scoped MVP claim. CoFiTok learns ordered denoising components on three public image settings: CIFAR-10, Tiny ImageNet-200, and an ImageNet-1K 64x64 fallback. Compared with an endpoint-only epsilon baseline trained for the same 20k steps, the K8 light denoise-path objective nearly matches final clean MSE on Tiny ImageNet-200 and ImageNet-64 HF, while dramatically improving prefix/path alignment and effective token usage. Generated-sample distribution smoke metrics do not show a stable unconditional generation-quality win, which clarifies the present contribution: CoFiTok makes dense noise prediction prefix-controllable without relying on a final VAE-style decoder.

Contributions.

1. We formulate dense pixel-space diffusion noise prediction as an ordered sum of denoising-token components.
2. We introduce a restricted, condition-free synthesis operator whose token-only interface and bias-free construction prevent decoder-style shortcuts and make $S_k(0) = 0$ testable.
3. We train CoFiTok with a denoise-path objective that makes token prefixes approximate a partial denoising trajectory rather than only the final clean image.
4. We evaluate prefix controllability, component ordering, non-degeneration, endpoint quality, and generated-sample smoke metrics across CIFAR-10, Tiny ImageNet-200, and ImageNet-64 HF.

2 Related Work

Diffusion models and latent diffusion. Denoising diffusion probabilistic models learn to reverse a Gaussian noising process by predicting noise targets [Ho et al., 2020]. Latent diffusion models move this process into a learned latent space and decode the final latent through a VAE-style decoder [Rombach et al., 2022]. CoFiTok instead stays in the pixel-space denoising algebra and factorizes the dense noise prediction itself. The CoFiTok token is not a final image latent; it is a negative-noise component that directly contributes to the pixel-space denoising update.

Trajectory ordering and latent-pixel interaction. Latent Forcing studies how changing the schedule between latent and pixel variables changes the information revealed by the diffusion trajectory [Baade et al., 2026]. CoFiTok is inspired by this view of trajectory ordering, but tokenizes the noise prediction inside the reverse trajectory rather than introducing a separate latent-pixel trajectory.

Ordered visual tokenizers. Spectral Image Tokenizer, FlexTok, Matryoshka Representation Learning, M3, and related nested-token approaches organize image or multimodal representations so that prefixes or subsets can be useful [Esteves et al., 2025, Bachmann et al., 2025, Kusupati et al., 2024, Cai et al., 2024]. These methods operate on image representations or visual-token budgets. CoFiTok operates on dense diffusion noise prediction. Its prefixes are valid pixel-space denoising updates, not partial decodes from a general image tokenizer.

Diffusion-autoregressive visual tokens. Selftok and D-AR are the closest conceptual neighbors because they connect diffusion processes and autoregressive tokens [Wang et al., 2025, Gao and Shou, 2025]. CoFiTok differs in the represented object and in the anti-degeneration mechanism. CoFiTok tokens are continuous negative-noise components, and the synthesis operator is deliberately weak and condition-free. The main claim is not that diffusion steps are visual tokens, but that dense noise prediction can be decomposed into ordered components whose prefixes are directly usable in the diffusion denoising formula.

3 Method

3.1 Forward Noising and Dense Prediction

For a clean image x_0 , timestep t , and Gaussian noise ϵ , the forward process is

$$x_t = \alpha_t x_0 + \sigma_t \epsilon. \quad (4)$$

A standard epsilon-prediction diffusion model learns $\epsilon_\theta(x_t, t, c)$. CoFiTok instead predicts the ordered component sum in Eq. 1. The current implementation uses unconditional models for the main MVP evidence. Class labels may exist in datasets, but the restricted synthesis operator never receives labels, prompts, timesteps, previous tokens, or image features. Figure 1 shows the dataflow and marks the blocked paths into S_k that define the default method boundary.

3.2 Next Denoising Tokens

The token predictor T_k may be expressive. It sees the noised image, timestep, and previous tokens:

$$z^{(1)} = T_1(x_t, t), \quad (5)$$

$$z^{(k)} = T_k(x_t, z^{(<k)}, t), \quad k > 1. \quad (6)$$

In the tiny MVP model, T_k is a small U-Net-like predictor with token heads. The simultaneous-prediction ablation disables token-to-token feedback so every token head reads the same hidden state.

3.3 Restricted Synthesis Operator

Each token is expanded independently:

$$\epsilon^{(k)} = S_k(z^{(k)}). \quad (7)$$

The default S_k is a bias-free projection followed by a bias-free local convolution. It is token-only and condition-free. Because no layer has bias and there is no learned constant input, zero tokens map to zero components:

$$S_k(0) = 0. \quad (8)$$

This restriction separates CoFiTok from a learned decoder. A deep- S_k ablation is included only to test degeneration risk: it uses biased nonlinear convolutions while preserving the token-only interface. This ablation can improve apparent endpoint metrics but fails the zero-token diagnostic, so it is not the default method.

CoFiTok method overview

Dense pixel-space noise prediction is factorized into ordered restricted denoising components.

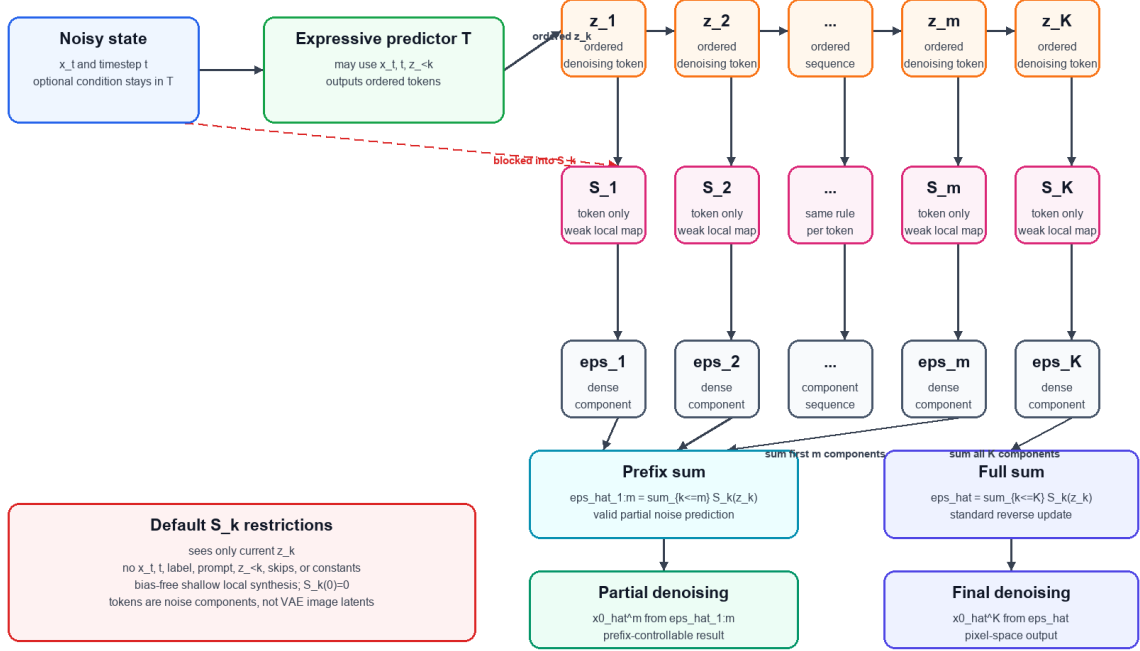


Figure 1: CoFiTok factorizes dense pixel-space diffusion noise prediction into ordered denoising tokens. The predictor may be expressive, but each synthesis operator S_k receives only the current token and is restricted to shallow, condition-free, bias-free local synthesis. Any prefix of components forms a valid partial noise prediction and therefore a valid pixel-space denoising result. Dashed blocked paths mark information that must not enter S_k , including x_t , timesteps, labels, prompts, previous tokens, skip features, and learned constants.

3.4 Prefix Denoising

For any prefix length m , CoFiTok forms the prefix prediction and pixel-space denoising result in Eq. 3. Thus every prefix is a valid partial noise prediction and every prefix produces a pixel-space denoising result. This is the primary interface exposed by the method.

3.5 Denoise-Path Objective

The current MVP objective is

$$\mathcal{L} = \mathcal{L}_\epsilon + 0.15 \mathcal{L}_{\text{path prefix}} + 0.30 \mathcal{L}_{\text{path component}}. \quad (9)$$

The denoise path interpolates from the zero-update x_t reconstruction toward coarse-to-fine spatial targets. The progress schedule uses power 1.5 in the current best MVP configuration. This objective deliberately avoids forcing early prefixes to be immediately clean; instead, prefixes are trained to follow a partial denoising path.

Table 1: Matched 20k two-seed comparison (mean $\pm$ sample standard deviation over seeds 103 and 139). Lower is better for final clean MSE and path AUC; higher is better for effective token count.

Dataset	Method	Final MSE $\downarrow$	Path AUC $\downarrow$	Eff. K $\uparrow$	Zero ratio $\downarrow$
Tiny ImageNet-200	Endpoint-only factorized	0.1253 $\pm$ 0.0001	1.0098 $\pm$ 0.1430	5.312 $\pm$ 0.049	0.0000 $\pm$ 0.0000
Tiny ImageNet-200	CoFiTok	0.1296 $\pm$ 0.0015	0.0378 $\pm$ 0.0007	6.766 $\pm$ 0.015	0.0000 $\pm$ 0.0000
ImageNet-64 HF	Endpoint-only factorized	0.1096 $\pm$ 0.0006	0.9984 $\pm$ 0.1210	5.169 $\pm$ 0.108	0.0000 $\pm$ 0.0000
ImageNet-64 HF	CoFiTok	0.1151 $\pm$ 0.0016	0.0379 $\pm$ 0.0037	6.752 $\pm$ 0.012	0.0000 $\pm$ 0.0000

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate on CIFAR-10 as a smoke and diagnostic dataset, Tiny ImageNet-200 as the main MVP dataset, and an ImageNet-1K 64x64 Hugging Face fallback as an ImageNet-family validation setting. The preferred TFDS `downsampled_imagenet/64x64` source was attempted but was not available in the current environment; therefore the fallback is always named explicitly.

Models. All current experiments use tiny models. The main method is K8 light denoise-path CoFiTok with restricted synthesis. We also evaluate K4 and K16 on ImageNet-64 HF, epsilon-only endpoint training, channel-mask controls, simultaneous-predictor ablations, and deep- S_k synthesis ablations.

Metrics. We report final clean MSE, prefix denoise-path AUC, effective token count, zero-token component energy ratio, shuffled-token relative mismatch, PSNR, LPIPS Alex, low-res Frechet proxy, torchvision Inception Frechet, and generated-sample smoke-to-50k streamed Frechet. Generated-sample Frechet is computed up to 8192 generated samples against 8192 real validation images for the main Tiny/HF rows, up to 50000/50000 for the HF ImageNet-64 fallback, and with matched 20k multiscale controls on Tiny and HF. This is still an internal torchvision Inception-Frechet protocol rather than an external official FID benchmark claim.

Validation. The current validation suite passes on the `pro6000` server. Summary artifacts contain 111 training reports, 95 order-evaluation reports, 58 fixed-timestep quality reports, 22 sampling reports, and 29 generated-quality reports. A deterministic full-validation reconstruction and prefix sweep covers the main Tiny ImageNet-200 10k-validation and ImageNet-64 HF 50k-validation rows. The stronger multiscale predictor also has matched 20k train-scale rows and generated-quality controls on Tiny ImageNet-200 and ImageNet-64 HF while preserving the restricted S_k contract.

4.2 Endpoint Quality and Prefixes

At the same 20k training budget, K8 light CoFiTok nearly matches epsilon-only final clean MSE while greatly improving path alignment.

Across the four paired dataset-seed comparisons, CoFiTok lowers path AUC in 4/4 cases, with a mean relative reduction of 96.31%. This structural gain has a measurable endpoint tradeoff: final MSE increases by 2.89% on average and is not lower in any of the four pairs. Figure 2 shows the corresponding qualitative prefix grids and synthesis diagnostics. In the full-validation deterministic sweep, K8 light also improves MSE AUC and low-res Frechet proxy AUC on both Tiny ImageNet-200 and ImageNet-64 HF, while endpoint-only factorization keeps the lower endpoint MSE.

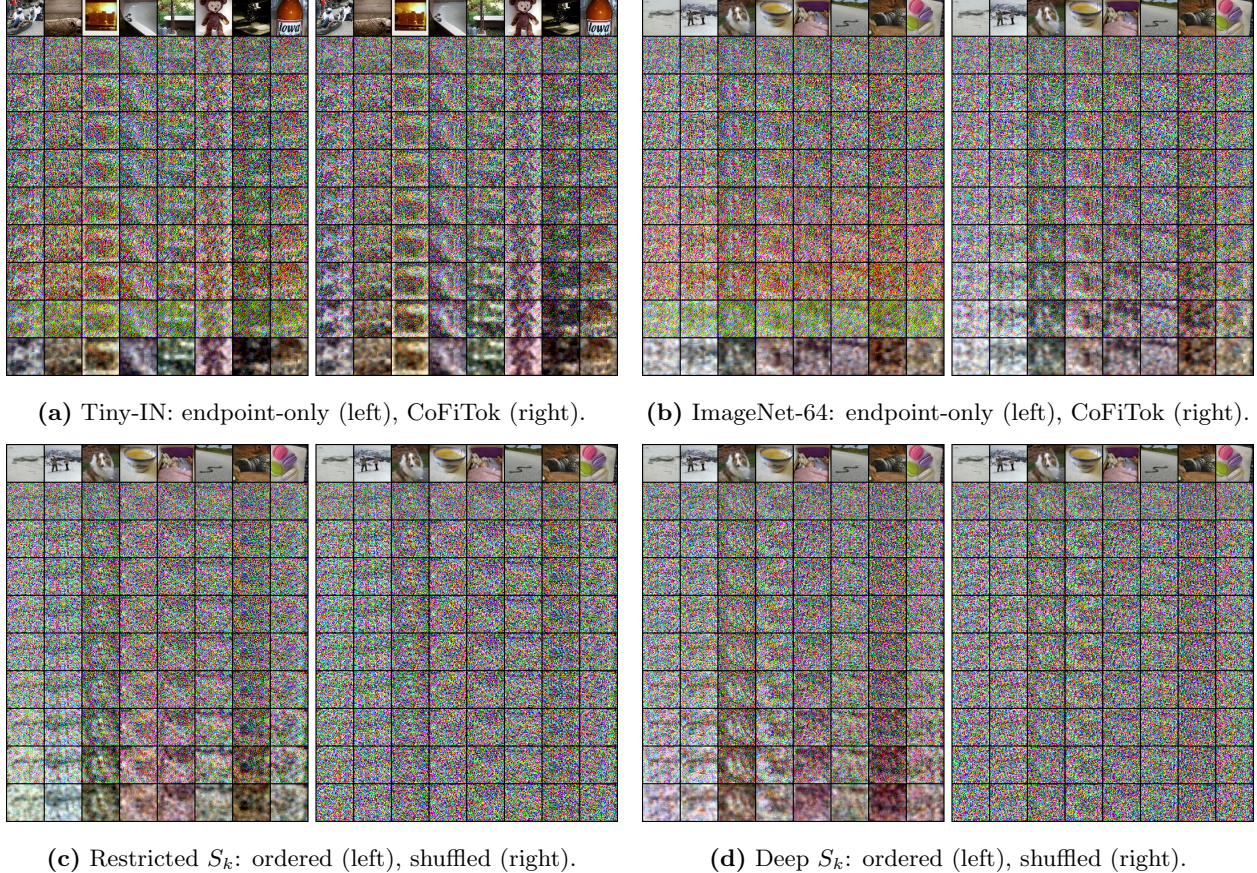


Figure 2: Qualitative prefix and degeneration evidence assembled from four standalone subfigure files. In (a–b), each half shows the same examples across the prefix path; CoFiTok produces more organized intermediate denoising than endpoint-only factorization. In (c–d), shuffling breaks sample alignment, while deep synthesis can carry structure that the restricted, zero-preserving operator cannot.

4.3 Order Sensitivity

Order ablations keep the same component sum at the endpoint but change the prefix trajectory. This makes order evaluation a direct test of whether prefixes are organized rather than an endpoint-quality test.

K8 gives the best current tradeoff: it uses tokens broadly and shows clear random/reverse degradation. K4 is a strong cost baseline but has less room for a rich prefix curriculum. K16 does not improve path AUC at 5k steps and likely requires longer training or stronger shaping before becoming worthwhile. Seed 103 repeats for K4 and K16 preserve the same pattern: reverse order damages prefix path quality in both cases, while K16 random order is also strongly worse than ordered accumulation.

4.4 Restricted Synthesis and Degeneration

The restricted synthesis operator has zero-token component energy ratio 0.0 in the restricted reports. The deep- S_k ablation improves some endpoint/path numbers but has a nonzero zero-token ratio: final clean MSE 0.0977, path AUC 0.0366, and zero ratio 0.0522. This result supports the

Table 2: ImageNet-64 HF order sensitivity at seed 139. The endpoint is unchanged by order because all components are still summed, but random and reverse orders damage prefix path quality, especially for K8/K16.

K	Order	Final MSE ↓	Path AUC ↓	Zero ratio ↓
4	ordered	0.1029	0.0674	0.0000
4	random	0.1029	0.0727	0.0000
4	reverse	0.1029	0.7516	0.0000
8	ordered	0.1041	0.0635	0.0000
8	random	0.1041	0.2441	0.0000
8	reverse	0.1041	0.6947	0.0000
16	ordered	0.1021	0.0680	0.0000
16	random	0.1021	0.4680	0.0000
16	reverse	0.1021	0.7020	0.0000

Table 3: Component-separation pressure ablation on Tiny ImageNet-200. Lower is better for ordered path AUC, mean absolute component cosine, and final MSE; higher is better for effective token count. Scheduled decor 0.003 consistently reduces component cosine while leaving ordered-prefix behavior intact.

Seed	Variant	Ordered AUC ↓	Mean abs cosine ↓	Final MSE ↓	Eff. K ↑
139	baseline	0.0766	0.7223	0.1487	6.8043
139	decor 0.003	0.0804	0.6890	0.1486	6.8316
103	baseline	0.0740	0.6967	0.1503	6.8430
103	decor 0.003	0.0776	0.6644	0.1501	6.8829

design boundary. A stronger synthesis operator can carry token-independent priors and improve apparent metrics, but then it no longer satisfies the default CoFiTok restriction. A seed-103 repeat shows the same failure mode with zero-token ratio 0.0450. Figure 2 visualizes restricted/deep and shuffled-token panels.

4.5 Component Separation Pressure

We treat component decorrelation as a pressure-term ablation rather than a default objective. The headline CoFiTok rows use the no-decorrelation light denoise-path objective because it most directly measures ordered prefix denoising. On Tiny ImageNet-200, a conservative schedule (**weight=0.003**, start step 3000, warmup 1500, full weight at step 4500 of 5000) reduces component correlation on two seeds while keeping the ordered-prefix penalty small.

Across the two seeds, scheduled decor 0.003 changes ordered path AUC by +0.0036, mean absolute component cosine by -0.0328, final MSE by -0.0002, and effective token count by +0.0336. Thus mild separation pressure can make components less correlated without materially breaking ordered prefix behavior. Stronger decor weights reduce cosine further but impose larger path-AUC penalties, so we keep them as stress-test ablations rather than the default CoFiTok objective.

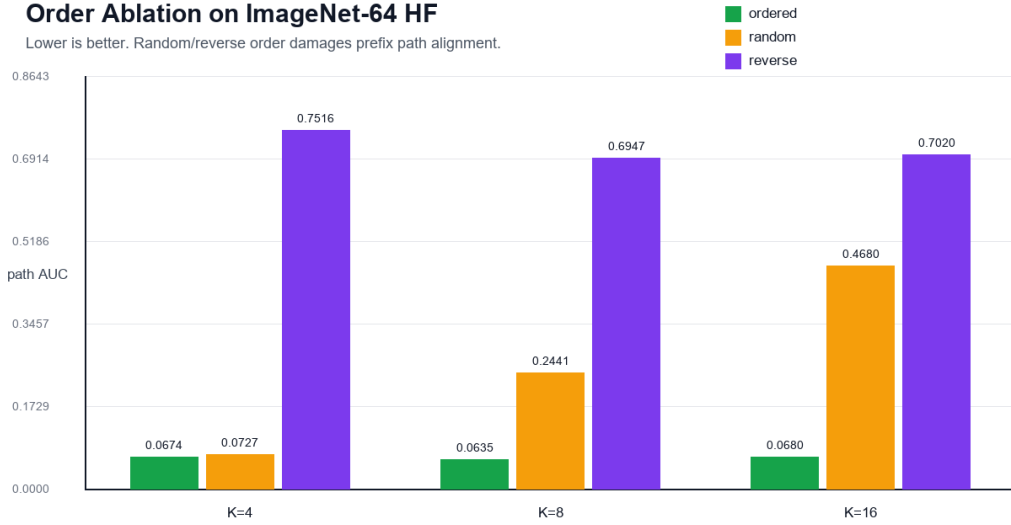


Figure 3: Component order affects prefix quality even when the final component sum is unchanged. Ordered components preserve the denoise path, while random and reverse orders damage prefix alignment.

Table 4: Cross-dataset fixed-timestep quality. These are reconstruction probes, not sample FID.

Dataset	Method	Steps	Final PSNR $\uparrow$	Final Inception $\downarrow$	Final LPIPS $\downarrow$
CIFAR-10	epsilon-only	3k	14.390	334.069	0.1639
CIFAR-10	K8 light	3k	14.170	327.889	0.1674
Tiny ImageNet-200	epsilon-only	20k	14.976	344.162	0.5479
Tiny ImageNet-200	K8 light	20k	14.865	346.083	0.5505
ImageNet-64 HF	epsilon-only	20k	15.743	330.643	0.5204
ImageNet-64 HF	K8 light	20k	15.558	360.795	0.5150

4.6 Fixed-Timestep and Generated-Sample Quality

Fixed-timestep 256-image validation shows that epsilon-only remains a strong endpoint baseline. K8 light is close on PSNR/LPIPS, but it is not uniformly better on endpoint metrics.

Generated-sample smoke metrics do not establish a stable unconditional generation-quality win at the current model scale. Earlier 1024/4096 DDIM-20 slices favor epsilon-only on both Tiny ImageNet-200 and ImageNet-64 HF; the 8192/8192 DDIM-50 streamed protocol slightly favors K8 light on Tiny ImageNet-200 but still favors epsilon-only on ImageNet-64 HF, and the 50000/50000 HF streamed protocol again favors epsilon-only. The matched 20k multiscale generated-quality controls also favor epsilon-only on both Tiny ImageNet-200 and ImageNet-64 HF under the internal streamed protocol, while the external `pytorch-fid` image-directory protocol is mixed.

Table 5: Generated-sample smoke-to-50k streamed metrics. Tiny ImageNet-200 is shown at 8192/8192 and with a matched multiscale 10000/10000 control; HF ImageNet-64 is shown at 8192/8192, 50000/50000, and matched multiscale 50000/50000. These are internal streamed DDIM Inception-Frechet measurements, not external official FID benchmark claims.

Dataset	Method	Generated	Real	Low-res Frechet ↓	Inception Frechet ↓
Tiny ImageNet-200	epsilon-only	8192	8192	7.5761	272.9955
Tiny ImageNet-200	K8 light	8192	8192	7.3264	271.0260
ImageNet-64 HF	epsilon-only	8192	8192	6.5490	240.8919
ImageNet-64 HF	K8 light	8192	8192	6.6993	269.2785
ImageNet-64 HF	epsilon-only	50000	50000	6.4496	238.0329
ImageNet-64 HF	K8 light	50000	50000	6.6096	266.8045
Tiny ImageNet-200	eps-only ms	10000	10000	5.2057	136.7132
Tiny ImageNet-200	K8 light ms	10000	10000	5.7001	141.4577
ImageNet-64 HF	eps-only ms	50000	50000	5.5593	134.2443
ImageNet-64 HF	K8 light ms	50000	50000	6.2099	136.8376

Table 6: Official `pytorch-fid` image-directory protocol for matched 20k multiscale checkpoints. Tiny ImageNet-200 is evaluated at 10000/10000 images and the ImageNet-64 HF fallback at 50000/50000 images. Lower is better. These rows are diagnostic evidence, not a stable generation-quality superiority claim.

Dataset	Method	Images	Official FID ↓
Tiny ImageNet-200	epsilon-only	10000	148.841
Tiny ImageNet-200	K8 light	10000	152.247
ImageNet-64 HF	epsilon-only	50000	140.721
ImageNet-64 HF	K8 light	50000	134.082

5 Discussion

CoFiTok suggests that dense pixel-space diffusion noise prediction can be treated as a sequence of ordered denoising-token components. The restricted synthesis operator is central: it gives the prefix interface meaning and prevents the method from becoming a generic latent decoder. The denoise-path objective is also central because early prefixes should represent partial denoising rather than clean reconstructions.

The strongest empirical result is the combination of near-matched final clean MSE and large prefix path-AUC gains at 20k steps. The most important negative result is that generated-sample Frechet smoke metrics do not show a stable unconditional generation-quality win. Together, these results define the current scope of the contribution.

6 Limitations

The current models are tiny and trained with MVP budgets. Generated-sample Frechet reaches 50000 generated vs 50000 real images for the HF fallback, and the external `pytorch-fid` image-directory protocol has been run for the matched multiscale 20k checkpoints. These sample-quality results are mixed and do not show a robust CoFiTok sample-quality win. Fixed-timestep Inception/LPIPS metrics use validation slices rather than full-dataset reconstruction sweeps. Seed repeats cover the main epsilon/K8 comparison but not every ablation. The ImageNet-family valida-

tion uses an HF ImageNet-1K 64x64 fallback because the preferred `downsampled_imagenet/64x64` source was unavailable in the current environment. The venue-neutral draft still needs author metadata and final checklist decisions before any camera-ready version.

7 Conclusion

CoFiTok factorizes pixel-space diffusion noise prediction into an ordered sequence of denoising tokens. Each token is expanded by a restricted, condition-free synthesis operator into a dense negative-noise component, and prefix sums yield valid partial denoising results. Across CIFAR-10, Tiny ImageNet-200, and ImageNet-64 HF, the current MVP evidence shows that CoFiTok can learn ordered, non-degenerate prefix-controllable components while preserving endpoint quality close to an epsilon-only baseline. The next step is to test whether the same factorization remains useful for larger backbones, longer training, and official generated-sample FID toolchains.

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